✅ Q1. Vehicle Management System (Layered Architecture)

📁 **Files:**

- model: Vehicle.java, ElectricVehicle.java, PetrolVehicle.java
- dao: VehicleDAO.java
- service: VehicleService.java
- ui: MainApp.java

📄 Vehicle.java

```
package model;
import java.util.Objects;

public abstract class Vehicle {
    protected int id;
    protected String name;
    protected double price;
    protected String status;

    public Vehicle(int id, String name, double price, String status) {
        this.id = id; this.name = name; this.price = price; this.status = status;
    }

    public int getId() { return id; }
    public String getName() { return name; }
    public double getPrice() { return price; }

    public void setPrice(double price) { this.price = price; }
    public void setStatus(String status) { this.status = status; }

    public abstract String getType();

    public boolean equals(Object o) {
        return o instanceof Vehicle && ((Vehicle)o).id == id;
    }
    public int hashCode() { return Objects.hash(id); }
```

```
    public String toString() {
        return id+" "+name+" "+price+" "+status+" "+getType();
    }
}
```

ElectricVehicle.java

```
package model;

public class ElectricVehicle extends Vehicle {
    int battery; double chargeTime;

    public ElectricVehicle(int id, String name, double price, String status, int b, double c) {
        super(id,name,price,status);
        battery=b; chargeTime=c;
    }
    public String getType(){ return "Electric"; }
}
```

PetrolVehicle.java

```
package model;

public class PetrolVehicle extends Vehicle {
    String fuel; double mileage;

    public PetrolVehicle(int id,String name,double price,String status,String f,double m){
        super(id,name,price,status);
        fuel=f; mileage=m;
    }
    public String getType(){ return "Petrol"; }
}
```

VehicleDAO.java

```java
package dao;
import model.Vehicle;
import java.util.*;

public class VehicleDAO {
   Set<Vehicle> set = new HashSet<>();

   public boolean add(Vehicle v){ return set.add(v); }
   public void remove(int id){ set.removeIf(v->v.getId()==id); }

   public Vehicle get(int id){
      return set.stream().filter(v->v.getId()==id).findFirst().orElse(null);
   }

   public Set<Vehicle> getAll(){ return set; }
}
```

VehicleService.java

```java
package service;
import dao.VehicleDAO;
import model.Vehicle;
import java.util.*;

public class VehicleService {
   VehicleDAO dao = new VehicleDAO();

   public void add(Vehicle v){ if(!dao.add(v)) System.out.println("Duplicate ID"); }
   public void remove(int id){ dao.remove(id); }

   public void updatePrice(int id,double p){
      Vehicle v=dao.get(id); if(v!=null) v.setPrice(p);
   }

   public void sell(int id){ Vehicle v=dao.get(id); if(v!=null) v.setStatus("sold"); }
   public void purchase(int id){ Vehicle v=dao.get(id); if(v!=null) v.setStatus("available"); }
```

```
    public void display(){ dao.getAll().forEach(System.out::println); }

    public void byType(String t){
        dao.getAll().stream().filter(v->v.getType().equalsIgnoreCase(t)).forEach(System.out::println);
    }

    public void byPrice(double min,double max){
        dao.getAll().stream().filter(v->v.getPrice()>=min &&
v.getPrice()<=max).forEach(System.out::println);
    }

    public void sort(){

dao.getAll().stream().sorted(Comparator.comparingDouble(Vehicle::getPrice)).forEach(System.o
ut::println);
    }

    public void search(String name){
        dao.getAll().stream().filter(v-
>v.getName().equalsIgnoreCase(name)).forEach(System.out::println);
    }
}
```

MainApp.java

```
package ui;
import model.*;
import service.VehicleService;
import java.util.*;

public class MainApp {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        VehicleService s=new VehicleService();

        while(true){
            System.out.println("1 Add 2 Remove 3 Update 4 Sell 5 Purchase 6 Display 7 Type 8 Price 9
```

```
Sort 10 Search 11 Exit");
        int ch=sc.nextInt();

        switch(ch){
          case 1:
            int type=sc.nextInt();
            int id=sc.nextInt();
            String name=sc.next();
            double price=sc.nextDouble();

            if(type==1)
              s.add(new ElectricVehicle(id,name,price,"available",100,2));
            else
              s.add(new PetrolVehicle(id,name,price,"available","Petrol",15));
            break;

          case 2: s.remove(sc.nextInt()); break;
          case 3: s.updatePrice(sc.nextInt(),sc.nextDouble()); break;
          case 4: s.sell(sc.nextInt()); break;
          case 5: s.purchase(sc.nextInt()); break;
          case 6: s.display(); break;
          case 7: s.byType(sc.next()); break;
          case 8: s.byPrice(sc.nextDouble(),sc.nextDouble()); break;
          case 9: s.sort(); break;
          case 10: s.search(sc.next()); break;
          case 11: System.exit(0);
        }
      }
    }
}
```

✅ **Q2–Q16 (COMMON TEMPLATE)**

👉 Use this **single reusable template** for ALL string questions

📄 **StringOperations.java**

```
import java.util.*;
import java.io.*;

public class StringOperations {
    public static void main(String[] args) throws Exception {
        Scanner sc = new Scanner(System.in);
        ArrayList<String> list = new ArrayList<>();

        while(true){
            String s=sc.nextLine();
            if(s.equals("stop")) break;
            list.add(s);
        }

        ArrayList<String> list2=new ArrayList<>();
        ArrayList<String> list3=new ArrayList<>();

        for(String s:list){

            // MODIFY CONDITIONS PER QUESTION

            if(s.length()>5) list2.add(s);

            if("aeiouAEIOU".indexOf(s.charAt(0))!=-1)
                list3.add(s);
        }

        System.out.println(list2);
        System.out.println(list3);

        write("file1.txt",list2);
        write("file2.txt",list3);
    }

    static void write(String f,List<String> l)throws Exception{
        BufferedWriter bw=new BufferedWriter(new FileWriter(f));
        for(String s:l) bw.write(s+"\n");
        bw.close();
```

```
    }
}
```

🔁 Replace Conditions Per Question

Q	list2 condition	list3 condition
Q2	length > 5	starts with vowel
Q4	length < 4	contains digit
Q6	consonant start	palindrome
Q8	uppercase	ends with vowel
Q10	first==last	only digits
Q12	contains "java"	uppercase
Q14	palindrome	contains space
Q16	repeated chars	no vowels

✅ Q17 – Longest Substring

```
import java.util.*;
class Q17{
 public static void main(String[] a){
  String s="abcabcbb";
  Set<Character> set=new HashSet<>();
  int i=0,j=0,max=0;

  while(j<s.length()){
   if(!set.contains(s.charAt(j))){
    set.add(s.charAt(j++));
    max=Math.max(max,set.size());
   } else set.remove(s.charAt(i++));
  }
  System.out.println(max);
```

```
 }
}
```

✅ Q18 – Longest Palindrome

```
class Q18{
 public static void main(String[] a){
  String s="babad",res="";
  for(int i=0;i<s.length();i++){
   String o=exp(s,i,i);
   String e=exp(s,i,i+1);
   if(o.length()>res.length()) res=o;
   if(e.length()>res.length()) res=e;
  }
  System.out.println(res);
 }
 static String exp(String s,int l,int r){
  while(l>=0&&r<s.length()&&s.charAt(l)==s.charAt(r)){l--;r++;}
  return s.substring(l+1,r);
 }
}
```

✅ Q19 – Product Except Self

```
class Q19{
 public static void main(String[] a){
  int[] nums={1,2,3,4};
  int n=nums.length;
  int[] res=new int[n];

  int left=1;
  for(int i=0;i<n;i++){
   res[i]=left;
   left*=nums[i];
  }

  int right=1;
```

```
  for(int i=n-1;i>=0;i--){
   res[i]*=right;
   right*=nums[i];
  }

  for(int x:res) System.out.print(x+" ");
 }
}
```

✅ Q20 – Rotate Matrix

```
class Q20{
 public static void main(String[] a){
  int[][] m={{1,2,3},{4,5,6},{7,8,9}};
  int n=m.length;

  for(int i=0;i<n;i++)
   for(int j=i;j<n;j++){
    int t=m[i][j]; m[i][j]=m[j][i]; m[j][i]=t;
   }

  for(int i=0;i<n;i++)
   for(int j=0;j<n/2;j++){
    int t=m[i][j];
    m[i][j]=m[i][n-1-j];
    m[i][n-1-j]=t;
   }

  for(int[] r:m){
   for(int x:r) System.out.print(x+" ");
   System.out.println();
  }
 }
}
```